JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

IIT JAM 2005 Official Paper

Category: Classic Era (2005–2014) • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
15	90 M	60 Min	MCQ

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	15	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2005 • Q1 • ID: JAM_2005_Q1) MCQ +6 M Neg: −2.0

1. Let $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ be sequences of real numbers such that $b_n = a_{2n}$ and $c_n = a_{2n+1}$. Then $\{a_n\}$ is convergent

(A) implies $\{b_n\}$ is convergent but $\{c_n\}$ need not be convergent

(B) implies $\{c_n\}$ is convergent but $\{b_n\}$ neet not be convergent

(C) implies both $\{b_n\}$ and $\{c_n\}$ are convergent

(D) if both $\{b_n\}$ and $\{c_n\}$ are convergent

Question 2 (JAM 2005 • Q2 • ID: JAM_2005_Q2) MCQ +6 M Neg: −2.0

2. An integrating factor of $x\dfrac{dy}{dx} + (3x+1)y = xe^{-2x}$ is

(A) $x\,e^{3x}$

(B) $3x\,e^{x}$

(C) $x\,e^{x}$

(D) $x^3 e^{x}$

Question 3 (JAM 2005 • Q3 • ID: JAM_2005_Q3) MCQ +6 M Neg: −2.0

3. The general solution of $x^2\dfrac{d^2y}{dx^2} - 5x\dfrac{dy}{dx} + 9y = 0$ is

(A) $(c_1 + c_2 x)e^{3x}$

(B) $(c_1 + c_2 \ln x)x^3$

(C) $(c_1 + c_2 x)x^3$

(D) $(c_1 + c_2 \ln x)e^{x^3}$

(Here c_1 and c_2 are arbitrary constants.)

Question 4 (JAM 2005 • Q4 • ID: JAM_2005_Q4) MCQ +6 M Neg: −2.0

4. Let $\vec{r} = x\,\hat{i} + y\,\hat{j} + z\,\hat{k}$. If $\phi(x,y,z)$ is a solution of the Laplace equation then the vector field $\left(\vec{\nabla}\phi + \vec{r}\right)$ is

(A) neither solenoidal nor irrotational

(B) solenoidal but not irrotational

(C) both solenoidal and irrotational

(D) irrotational but not solenoidal

Question 5 (JAM 2005 • Q5 • ID: JAM_2005_Q5) MCQ +6 M Neg: −2.0

5. Let $\vec{F} = x\,\hat{i} + 2y\,\hat{j} + 3z\,\hat{k}$, S be the surface of the sphere $x^2 + y^2 + z^2 = 1$ and $\hat{n}$ be the inward unit normal vector to S. Then $\iint_S \vec{F} \cdot \hat{n}\, dS$ is equal to

(A) 4π

(B) -4π

(C) 8π

(D) -8π

Question 6 (JAM 2005 • Q6 • ID: JAM_2005_Q6) MCQ +6 M Neg: −2.0

6. Let A be a 3×3 matrix with eigenvalues 1, -1 and 3. Then

(A) $A^2 + A$ is non-singular

(B) $A^2 - A$ is non-singular

(C) $A^2 + 3A$ is non-singular

(D) $A^2 - 3A$ is non-singular

Question 7 (JAM 2005 • Q7 • ID: JAM_2005_Q7) MCQ +6 M Neg: −2.0

7. Let $T : \boldsymbol{R}^3 \longrightarrow \boldsymbol{R}^3$ be a linear transformation and I be the identity transformation of $\boldsymbol{R}^3$. If there is a scalar c and a non-zero vector $x \in \boldsymbol{R}^3$ such that $T(x) = cx$, then rank$(T - cI)$

(A) cannot be 0

(B) cannot be 1

(C) cannot be 2

(D) cannot be 3

Question 8 (JAM 2005 • Q8 • ID: JAM_2005_Q8) MCQ +6 M Neg: −2.0

8. In the group $\{1, 2, \ldots, 16\}$ under the operation of multiplication modulo 17, the order of the element 3 is

(A) 4

(B) 8

(C) 12

(D) 16

Question 9 (JAM 2005 • Q9 • ID: JAM_2005_Q9) MCQ +6 M Neg: −2.0

9. A ring R has maximal ideals

(A) if R is infinite

(B) if R is finite

(C) if R is finite with at least 2 elements

(D) only if R is finite

Question 10 (JAM 2005 • Q10 • ID: JAM_2005_Q10) MCQ +6 M Neg: −2.0

10. The integral $\int_0^1 \left[\int_0^{1-z} \left(\int_0^2 dx\right) dy\right] dz$ is equal to

(A) $\int_0^1 \left[\int_0^{1-y} \left(\int_0^2 dx\right) dz\right] dy$

(B) $\int_0^1 \left[\int_0^{1-y} \left(\int_0^1 dx\right) dz\right] dy$

(C) $\int_0^2 \left[\int_0^2 \left(\int_0^{1-z} dx\right) dz\right] dy$

(D) $\int_0^2 \left[\int_0^2 \left(\int_0^{1-y} dx\right) dz\right] dy$

Question 11 (JAM 2005 • Q11 • ID: JAM_2005_Q11) MCQ +6 M Neg: −2.0

11. Let $f : \boldsymbol{R} \longrightarrow \boldsymbol{R}$ be continuous and $g, h : \boldsymbol{R}^2 \longrightarrow \boldsymbol{R}$ be differentiable. Let $F(u,v) = \int_v^u f(t)dt$, where $u = g(x,y)$ and $v = h(x,y)$. Then $\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} =$

(A) $f(g(x,y))\left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y}\right] - f(h(x,y))\left[\frac{\partial h}{\partial x} + \frac{\partial h}{\partial y}\right]$

(B) $f(h(x,y))\left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y}\right] - f(g(x,y))\left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y}\right]$

(C) $f(h(x,y))\left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y}\right] + f(g(x,y))\left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y}\right]$

(D) $f(g(x,y))\left[\frac{\partial g}{\partial x} - \frac{\partial g}{\partial y}\right] + f(g(x,y))\left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y}\right]$

Question 12 (JAM 2005 • Q12 • ID: JAM_2005_Q12) MCQ +6 M Neg: −2.0

12. Let $y = f(x)$ be a smooth curve such that $0 < f(x) < K$ for all $x \in [a,b]$. Let

L = length of the curve between $x = a$ and $x = b$

A = area bounded by the curve, x-axis, and the lines $x = a$ and $x = b$

S = area of the surface generated by revolving the curve about x-axis between $x = a$ and $x = b$

Then

(A) $2\pi KL < S < 2\pi A$

(B) $S \le 2\pi A < 2\pi KL$

(C) $2\pi A \le S < 2\pi KL$

(D) $2\pi A < 2\pi KL < S$

Question 13 (JAM 2005 • Q13 • ID: JAM_2005_Q13) MCQ +6 M Neg: −2.0

13. Let $f : \boldsymbol{R} \longrightarrow \boldsymbol{R}$ be defined by $f(t) = t^2$ and let U be any non-empty open subset of $\boldsymbol{R}$. Then

(A) $f(U)$ is open

(B) $f^{-1}(U)$ is open

(C) $f(U)$ is closed

(D) $f^{-1}(U)$ is closed

Question 14 (JAM 2005 • Q14 • ID: JAM_2005_Q14) MCQ +6 M Neg: −2.0

14. Let $f : (-1,1) \longrightarrow \boldsymbol{R}$ be such that $f^{(n)}(x)$ exists and $\left|f^{(n)}(x)\right| \le 1$ for every $n \ge 1$ and for every $x \in (-1,1)$. Then f has a convergent power series expansion in a neighbourhood of

(A) every $x \in (-1,1)$

(B) every $x \in \left(\frac{-1}{2}, 0\right)$ only

(C) no $x \in (-1,1)$

(D) every $x \in \left(0, \frac{1}{2}\right)$ only

Question 15 (JAM 2005 • Q15 • ID: JAM_2005_Q15) MCQ +6 M Neg: −2.0

15. Let $a > 1$ and $f, g, h : [-a, a] \longrightarrow \boldsymbol{R}$ be twice differentiable functions such that for some c with $0 < c < 1 < a$,

$$f(x) = 0 \text{ only for } x = -a, 0, a;$$
$$f'(x) = 0 = g(x) \text{ only } x = -1, 0, 1;$$
$$g'(x) = 0 = h(x) \text{ only for } x = -c, c.$$

The possible relations between f, g, h are

(A) $f = g'$ and $h = f'$

(B) $f' = g$ and $g' = h$

(C) $f = -g'$ and $h' = g$

(D) $f = -g'$ and $h' = f$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2005 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+6	**C**
2	MCQ	+6	**A**
3	MCQ	+6	**B**
4	MCQ	+6	**D**
5	MCQ	+6	**D**
6	MCQ	+6	**C**
7	MCQ	+6	**D**
8	MCQ	+6	**D**

Q. No.	Type	Marks	Official Key
9	MCQ	+6	**C**
10	MCQ	+6	**A**
11	MCQ	+6	**A**
12	MCQ	+6	**C**
13	MCQ	+6	**B**
14	MCQ	+6	**A**
15	MCQ	+6	**B**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.